



King Saud University

College of Business Administration

Management Information Systems Department

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“The Financial Challenges Faced by Canadian Households During the COVID-19
Pandemic”

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Introduction to power BI

Power BI is a data visualization and business intelligence tool by Microsoft. It helps users connect to various data sources, transform raw data , and create interactive dashboards and reports , With its easy-to-use interface, Power BI allows individuals and organizations to analyze data, track performance, and make informed decisions through clear visual insights

Data Description

The following table describes the columns in the dataset used for analyzing the impact of COVID-19 on the consumption patterns of Canadian families. It includes the column name, data type, and a brief description of each variable

Column Name	Data Type	Description
Family Size	Integer	Number of family members
Age Group	String	Age range of the family (e.g., 20-30 years)
Area	String	Type of residential area (Urban, Suburban, Rural)
Income-Before Covid	Float	Family's monthly income before the COVID-19 pandemic
Income -After Covid	Float	Family's monthly income after the COVID-19 pandemic
Expenses-Before Covid	Float	Family's monthly expenses before the COVID-19 pandemic
Expenses-After Covid	Float	Family's monthly expenses after the COVID-19 pandemic

We were able to analyze the changes in family consumption patterns before and after the COVID-19 pandemic by using specific columns,

namely (Family Size, Area, Income (Before Covid), Income (AfterCovid), Expenses (Before Covid), and Expenses (After Covid)).

This data and its role are crucial for us to accurately identify the impact of the pandemic, and it supports decision-makers in creating strategies that enhance financial stability for families.

Dataset Screenshot

REF_DATE	GEO	Statistic	Before-tax household income quintile	Household expenditures, summary-level categories
2011	Province 1	Average expenditure per household	Fourth quintile	Owned living quarters
2011	Province 1	Average expenditure per household	Fourth quintile	Principal accommodation
2011	Province 1	Average expenditure per household	Fourth quintile	Shelter
2011	Province 1	Average expenditure per household	Fourth quintile	Transportation
2011	Province 1	Average expenditure per household	Fourth quintile	Food purchased from stores
2011	Province 1	Average expenditure per household	Fourth quintile	Total expenditure
2011	Province 1	Average expenditure per household	Fourth quintile	Communications
2011	Province 1	Average expenditure per household	Fourth quintile	Food purchased from restaurants
2011	Province 1	Average expenditure per household	Fourth quintile	Rented living quarters
2011	Province 1	Average expenditure per household	Fourth quintile	Water, fuel and electricity for principal accommodation
2011	Province 1	Average expenditure per household	Fourth quintile	Other accommodation
2011	Province 1	Average expenditure per household	Fourth quintile	Household equipment
2011	Province 1	Average expenditure per household	Fourth quintile	Household appliances
2011	Province 1	Average expenditure per household	Fourth quintile	Household furnishings
2011	Province 1	Average expenditure per household	Fourth quintile	Household operations
2011	Province 1	Average expenditure per household	Fourth quintile	Clothing and accessories
2011	Province 1	Average expenditure per household	Fourth quintile	Household furnishings and equipment
2011	Province 10	Average expenditure per household	Fourth quintile	Clothing and accessories
2011	Province 10	Average expenditure per household	Fourth quintile	Household operations
2011	Province 10	Average expenditure per household	Fourth quintile	Owned living quarters
2011	Province 10	Average expenditure per household	Fourth quintile	Shelter
2011	Province 10	Average expenditure per household	Fourth quintile	Principal accommodation
2011	Province 10	Average expenditure per household	Fourth quintile	Other accommodation
2011	Province 10	Average expenditure per household	Fourth quintile	Household equipment

Expense	Family type	Age of older adult	Family income	Income
11279	Couple families	45 to 54 years	\$30,000 and over	1718330
16095	Couple families	45 to 54 years	\$20,000 and over	1789860
17525	Couple families	45 to 54 years	\$15,000 and over	1814950
14183	Couple families	45 to 54 years	Median total income	100600
6598	Couple families	45 to 54 years	Under \$10,000	40660
86821	Couple families	45 to 54 years	All income groups	1873330
2091	Couple families	45 to 54 years	\$50,000 and over	1533200
2547	Couple families	45 to 54 years	\$10,000 and over	1832670
2136	Couple families	45 to 54 years	\$25,000 and over	1756020
2679	Couple families	45 to 54 years	\$35,000 and over	1677260
1430	Couple families	45 to 54 years	\$40,000 and over	1632310
1269	Couple families	45 to 54 years	\$75,000 and over	1250910
553	Couple families	45 to 54 years	\$80,000 and over	1189660
974	Couple families	45 to 54 years	\$70,000 and over	1310720
4989	Couple families	45 to 54 years	\$45,000 and over	1583850
4004	Couple families	45 to 54 years	\$90,000 and over	1065590
2378	Couple families	45 to 54 years	\$60,000 and over	1425860
3442	Couple families	45 to 54 years	\$90,000 and over	5130
4512	Couple families	45 to 54 years	\$45,000 and over	7760
10182	Couple families	45 to 54 years	\$30,000 and over	8130
15674	Couple families	45 to 54 years	\$15,000 and over	8370
14446	Couple families	45 to 54 years	\$20,000 and over	8280
1227	Couple families	45 to 54 years	\$40,000 and over	7920
1136	Couple families	45 to 54 years	\$75,000 and over	6150

Problem or opportunity formulation

COVID-19 Financial Impact on Canadian Families: An Analytical Story

Introduction to the Problem (Story)

The COVID-19 pandemic was not just a health emergency; it became a pivotal moment that reshaped how Canadian families lived, worked, and handled their finances. Households across the nation were suddenly facing job losses, declining incomes, and overwhelming uncertainty. As a result, families had to make tough financial choices, adjusting their spending habits based on necessity and adaptability.

Throughout 2020 and 2021, Canadians spent significantly less on non-essential items like travel, recreation, and transportation, while channeling more resources toward essentials such as housing, food, and healthcare. This shift was not solely about survival—it indicated a deeper change in values, with many prioritizing basic needs over luxuries and often choosing security over comfort.

This trend was especially pronounced in urban areas, where stricter lockdowns and higher living costs intensified financial pressures. Older Canadians, particularly those aged 45 to 64, were among the hardest hit as they balanced income loss, caregiving responsibilities, and future financial planning. Meanwhile, younger adults showed reduced financial activity, likely due to unstable employment opportunities during the crisis.

Our dashboard vividly captures this story. Each data point represents how Canadian families navigated uncertainty, made difficult decisions, and adapted their daily lives. It tells a broader story of resilience, adaptability, and the deeply human response to an unprecedented event.

Problem Approach

The COVID-19 pandemic triggered urgent and widespread financial disruption for Canadian households. To tackle this, we collected real-world data highlighting changes in consumer spending, income stability, and demographic behavior throughout 2020 and 2021. By analyzing these patterns, we aimed to uncover hidden financial vulnerabilities and adaptive strategies across different population groups.

Data Preparation and Problem Formulation

Before diving into analysis, we gathered raw data from multiple sources, including online datasets, legacy systems, and social data. Our goal was to create a comprehensive overview of household financial behavior during the pandemic.

Next, we focused on problem formulation, clearly defining our objective: to understand how Canadian families shifted their spending priorities during COVID-19 and to identify the most vulnerable demographic groups. This clarity provided a solid foundation and direction for the entire project.

Data Cleaning Process (Detailed Steps)

Ensuring the accuracy and reliability of our dataset was crucial. Our detailed cleaning steps included

Checking for missing values: We identified gaps in key fields like age, income, and expenses. Where possible, we filled in missing values using averages for specific age groups or logical estimates. Rows with excessive missing data were removed

Removing duplicates: We eliminated repeated entries to ensure each household was counted only once

Standardizing information: We unified inconsistent entries, such as different spellings for regions or variations in age group categories, ensuring consistency across the dataset

Handling outliers: Unrealistic income or expense values were flagged and either corrected or removed to prevent skewed analysis results

Balancing samples: We ensured fair representation across demographic groups (by age and geography) for unbiased comparisons

Final validation: After cleaning, we reviewed totals and preliminary visualizations to confirm that the data made logical sense before moving on to deeper analysis

Thanks to these structured steps, we created a robust, reliable foundation for the remainder of our project

Affected Parties and Impact

Certain groups were especially impacted

Middle-aged Canadians (ages 45–64): Faced increased financial strain, balancing job losses, caregiving responsibilities, and retirement planning

Young adults (ages 18–30): Dealt with unstable employment and reduced economic activity, potentially affecting their long-term financial stability

Wider consequences included rising household debt, decreased personal savings, and widening economic disparities among different demographic groups

Benefits of Solving the Problem

Addressing these financial vulnerabilities can lead to

Stronger, more responsive social safety nets crafted by policymakers

Financial products designed to better withstand income shocks, developed by financial institutions

Improved financial education and preparedness for families, contributing to greater •
.overall economic stability

Visualization Tools Used

:We utilized Power BI to develop a dashboard that visualizes

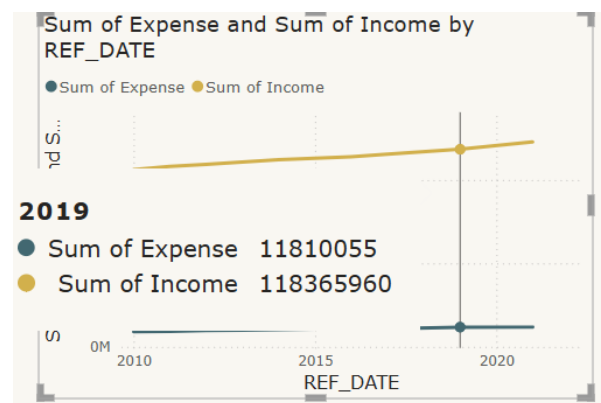
.Shifts in spending across different categories before and during the pandemic •

.Variations in income among different demographic groups •

.Geographic differences between urban and rural households •

These visual tools helped us identify key trends, highlight vulnerable groups, and

.reinforce our analysis with clear, data-driven insights



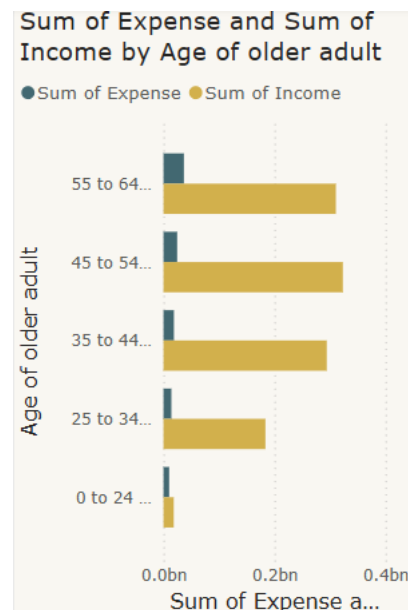
Line Chart by “REF_DATE” (Date Reference):

Reason for Choosing:

A line chart is the best choice for showing trends over time. It helps illustrate changes in income and expenses over the years, making it great for showing the impact of COVID-19.

Explanation: This chart shows income and expenses over time (from 2010 to 2020).

Link to COVID-19: The chart clearly shows the impact of COVID-19 around 2020, highlighting the drop in income and increase in expenses during the pandemic.



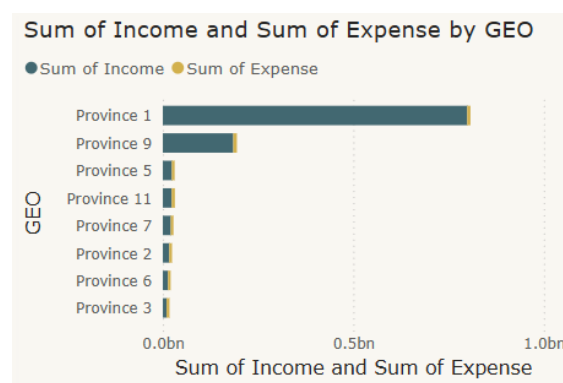
Bar Chart by “Age of Older Adult”:

Reason for Choice:

This chart helps compare income and expenses across different age groups. Age groups are important for understanding how spending behavior differs across age ranges.

This chart shows how older adults’ spending might have been impacted differently during the pandemic.

Explanation: The chart compares income and expenses across age groups such as 25-34 years, 35-44 years, etc. Link to COVID-19: This chart shows how older age groups were likely affected by the pandemic, especially with increased spending on healthcare due to the pandemic.



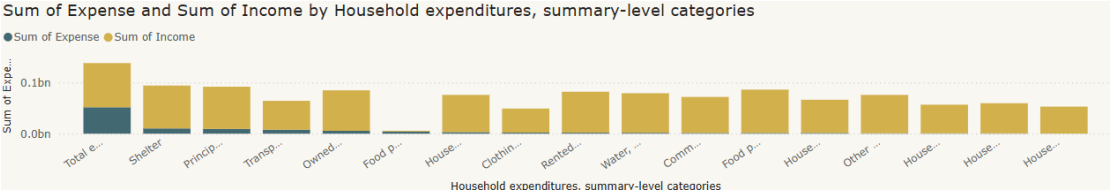
Bar Chart by “GEO” (Geographical Regions):

Reason for Choice: This chart is useful for comparing income and expenses across different geographical regions (or provinces) in Canada.

It allows for comparison of how different regions were affected by the pandemic.

Explanation: This chart compares income and expenses across different provinces (e.g., Province 1, Province 2, etc.).

Link to COVID-19: It shows how different provinces in Canada were impacted differently by the pandemic, with some regions likely experiencing higher income losses or higher healthcare costs due to COVID-19.



Stacked Column Chart by “Household Expenditures Summary-Level Categories”:

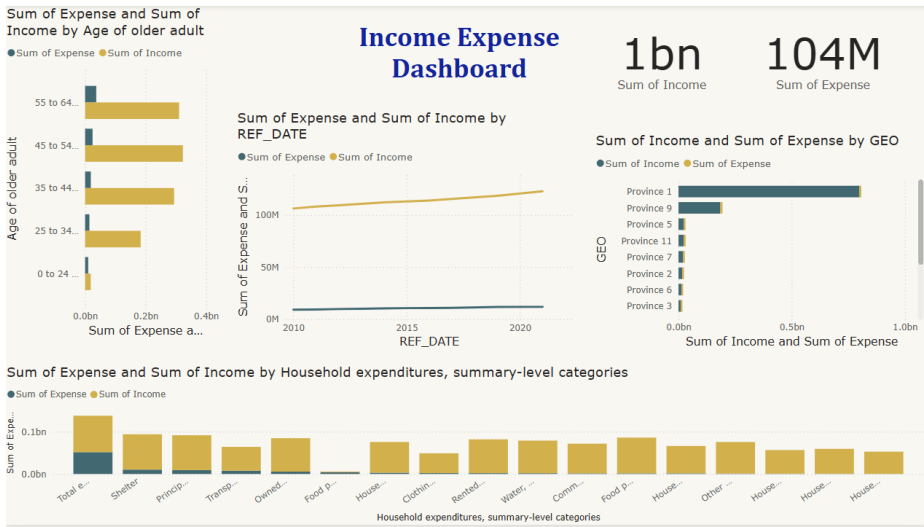
Reason for Choice: This chart helps display household spending across various categories (such as Rent, Food, Healthcare, etc.).

It gives a complete view of how spending patterns have shifted across different categories over time.

Explanation: This chart breaks down household expenditures into various categories like Shelter, Food, Transport, and others.

Link to COVID-19: It can show how spending priorities have shifted due to the pandemic, such as increased spending on healthcare or food due to lockdowns and price hikes.

Data Visualization



Link Dashboard in power BI

[Click Here](#)

Solution

To address the financial challenges faced by Canadian households during the COVID-19 pandemic, it is essential to provide targeted financial support to the most affected groups, particularly low-income families and urban residents. This support could include direct cash assistance or programs to ease the burden of basic living expenses. In addition, offering housing facilitation measures, such as rent reductions, mortgage payment deferrals, or financial assistance for housing costs, is critical given that shelter was one of the largest household expenses during the pandemic. Supporting employment and workforce participation is also vital by establishing programs for retraining unemployed individuals and promoting flexible job opportunities and remote work. These initiatives would help families regain financial stability and continue planning for a more secure future.

Conclusion

The COVID-19 pandemic dramatically reshaped the financial landscape for Canadian households, revealing both vulnerabilities and resilience across different demographics. Through the analysis of key data points—family size, area of residence, income, and expenses before and after the pandemic—we observed a clear shift in consumption patterns: a prioritization of essential needs over discretionary spending. Urban families and individuals aged 45–64 faced the greatest financial strain, grappling with higher living costs and income instability.

These findings highlight the importance of targeted interventions, such as direct financial support, housing cost relief, and employment retraining programs. By using data-driven insights, decision-makers can craft more effective strategies to support households in recovery and build economic resilience against future crises. Ultimately, the story behind the numbers shows not just financial adaptation, but also a profound reorientation of values—emphasizing security, essential well-being, and future preparedness among Canadian families.

References

[1] Statistics Canada, "Impact of COVID-19 on households in Canada," Statistics Canada, [Online]. Available: <https://www.statcan.gc.ca>. [Accessed: Apr. 24, 2025].

[2] Government of Canada, "COVID-19 financial support measures," Government of Canada, [Online]. Available: <https://www.canada.ca/en/department-finance/economic-response-plan.html>. [Accessed: Apr. 24, 2025].

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